



SIGNAL

System for Intelligent Geopolitical News and Asset Linking
User Operation Manual

Shell Energy — Enterprise Crude Oil Market Intelligence Platform
Version 2.0 | Analyst & Executive Edition | Confidential



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1. Introduction & Platform Overview

SIGNAL (System for Intelligent Geopolitical News and Asset Linking) is Shell's enterprise-grade crude oil market intelligence platform. It fuses real-time news sentiment analysis with quantitative price modelling to give traders and executives a unified decision-support environment. This manual covers every feature available in SIGNAL v2 and is intended for both Analyst and Executive users.

Feature	Description
Real-Time News Ingestion	Automated scraping & NLP tagging of crude-market headlines
Directional AI Model	81.5 % accurate UP/DOWN classifier (Random Forest + XGBoost)
Price Forecast Model	Stacking ensemble — RMSE 2.07 vs baseline 2.09; R ² 0.9532
Command Center	Live KPI summary with holdout chart, price chart, alerts
Commodity View	WTI/Brent historical & live price charts with technicals
Analytics	Sentiment trend, article volume, signal confidence heat-maps
Settings	Per-user alerts, themes, session management

2. User Roles & Access Control

SIGNAL implements role-based access control. Permissions are stored client-side and enforced by the sidebar navigation renderer. Two roles are currently supported:

Role	Accessible Pages	Badge
Analyst	Command Center, Intelligence Feed, Commodity View, Analytics, Direction Model, Price-Forecast, Settings	
Executive	All Analyst pages + Executive-only briefing flows during onboarding	Gold EXEC badge in sidebar

3. Getting Started — Login & Registration

SIGNAL uses a local authentication system. User credentials are stored in the browser's localStorage. No server-side authentication is required for the frontend, making the platform portable and deployable without a dedicated auth service.

Login

Navigate to the root URL (/) to reach the SIGNAL login page. Enter your registered email address and password, then click **Sign In**.

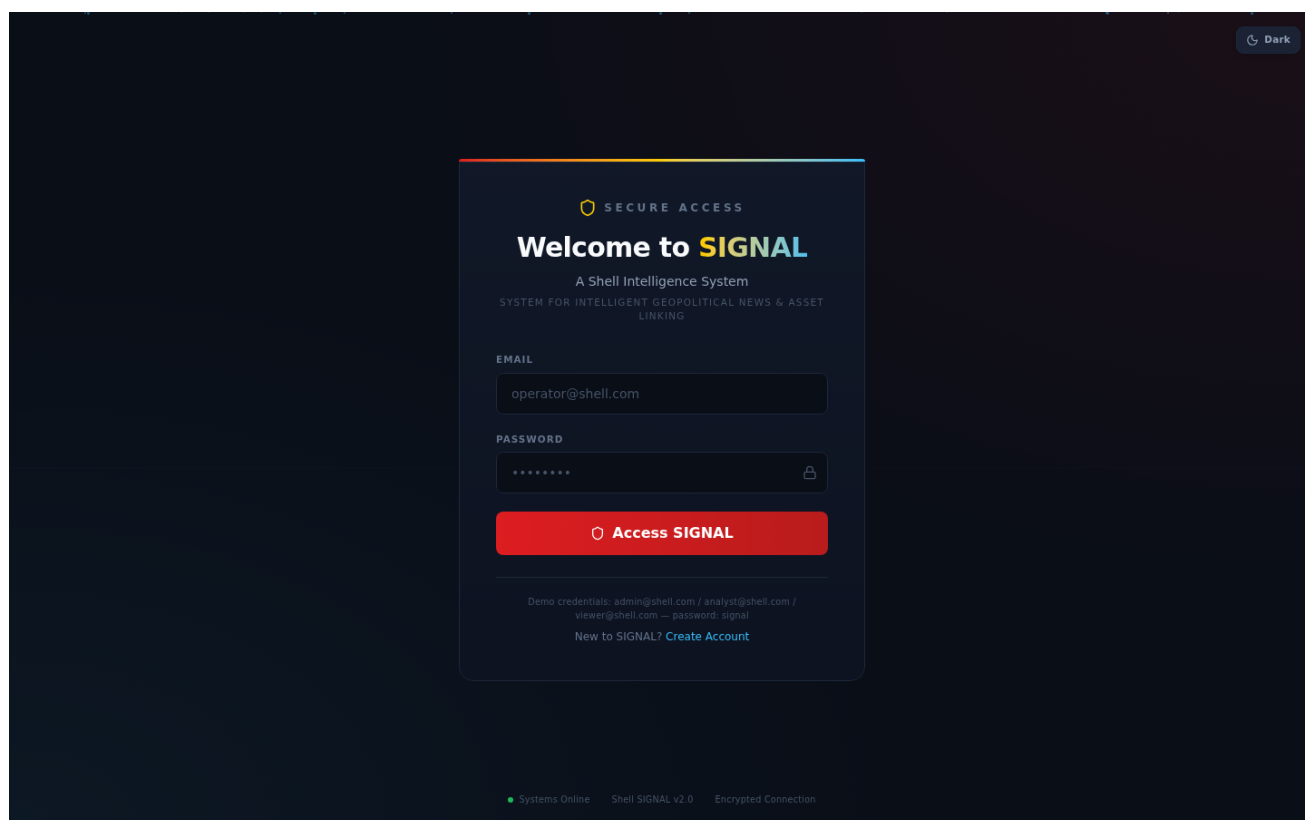


Fig 3-1 — SIGNAL Login Page

Registration

New users click **Create an Account** on the login page. On the Register page:

- Enter your full name, email address, and choose a password.
- Select your role: **Analyst** or **Executive**.
- Click **Create Account**. You will be redirected to the Onboarding flow.

Dark

CREATE ACCOUNT

Join SIGNAL

SHELL INTELLIGENCE SYSTEM REGISTRATION

FULL NAME

Alexandra Chen

SHELL EMAIL

name@shell.com

PASSWORD

CONFIRM

ACCESS CODE (OPTIONAL)

Enter access code for elevated privileges

Create Account

Already have an account? [Sign in](#)

Systems Online

Shell SIGNAL V2.0

Encrypted Connection

Fig 3-2 — SIGNAL Registration Page

Note: Re-registering with the same email will reset your onboarding state and alert configuration.

4. Onboarding Flow

All new users complete a guided onboarding sequence before accessing the dashboard. The flow differs based on role.

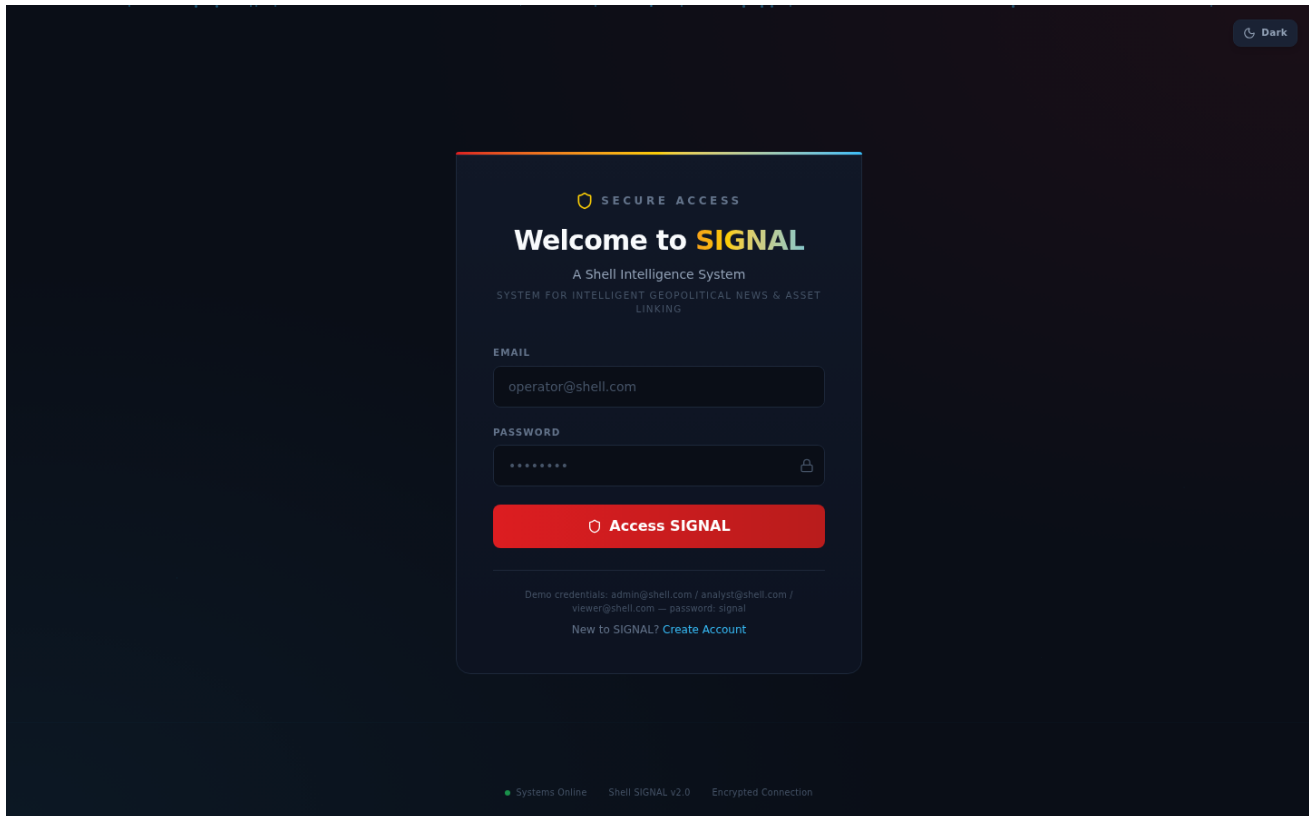


Fig 4-1 — Onboarding Flow (initial screen)

4.1 Analyst Onboarding (5 Steps)

- Step 1: Welcome & overview
- Step 2: Alert configuration — set commodity thresholds, sentiment levels, and notification preferences
- Step 3: Data source acknowledgement
- Step 4: Dashboard preview
- Step 5: Activation — access granted

4.2 Executive Onboarding (6 Steps)

- Step 1: Cinematic security verification (Phase 0) — visual identity confirmation animation
- Step 2: Strategic briefing — live KPI data pulled from the platform
- Step 3: Strategic focus areas — select priority commodities and regions
- Step 4: Escalation protocols — set briefing cadence and escalation level
- Step 5: Dashboard preview
- Step 6: Activation sequence — EXEC badge applied to sidebar

Note: Onboarding can only be replayed by re-registering the account. Alert configurations are editable post-onboarding in Settings.

5. Command Center (/dashboard)

The Command Center is the platform's default landing page after login. It provides a high-level situational-awareness summary: live prices, AI signal, model performance, and recent news — all on one screen.

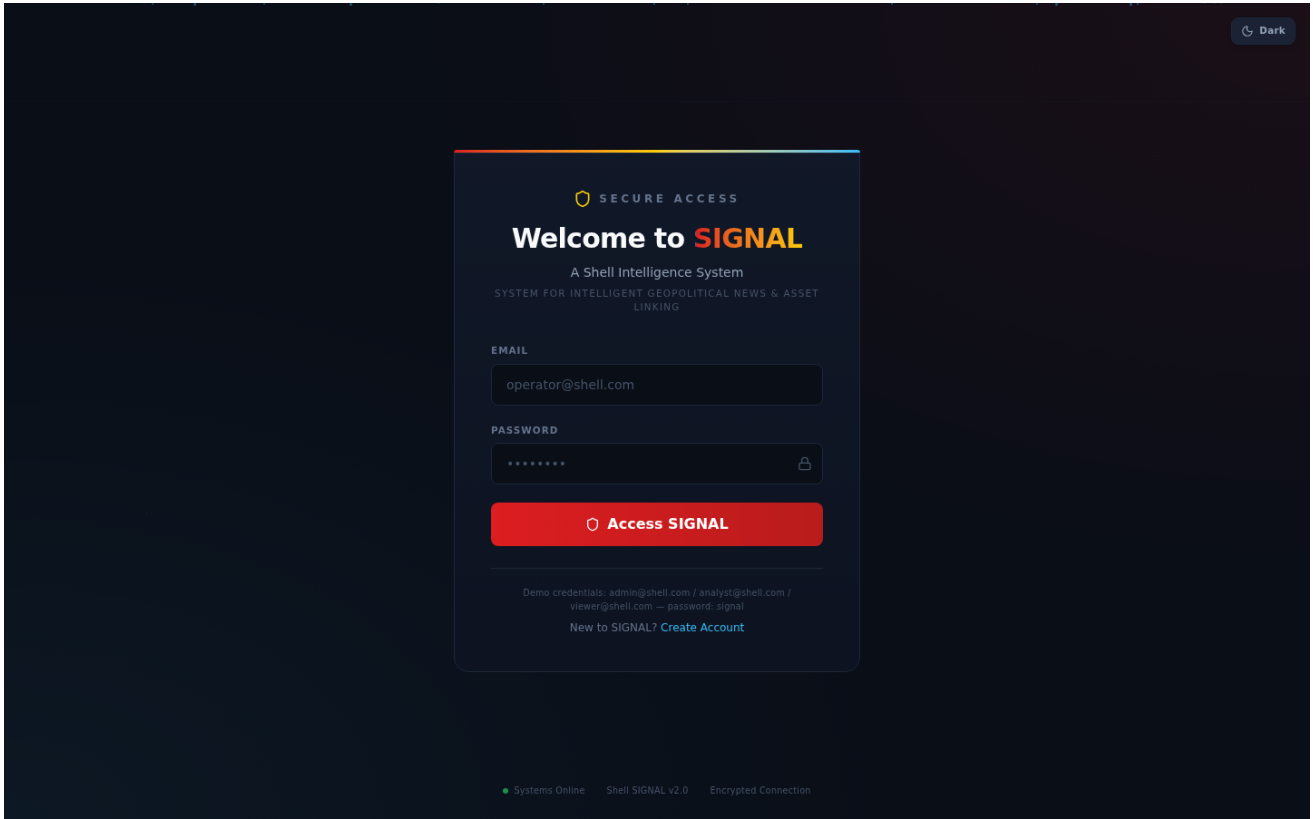


Fig 5-1 — Command Center Dashboard

5.1 KPI Cards (top row)

Card	Description
WTI Price	Latest West Texas Intermediate spot price in \$/bbl
Today's Signal	AI directional prediction (UP ▲ / DOWN ▼) with confidence %
24h Change	Absolute and percentage change since previous close
Sentiment Score	Aggregated news sentiment from the last 24 hours (–1 to +1)

5.2 Holdout Chart Panel (■ width)

Displays the model's actual vs. predicted price comparison over the holdout test set, alongside a naïve random-walk baseline. Three-line chart: **Actual** (white), **Predicted** (blue), **Baseline** (orange).

5.3 Price Chart Panel (■ width)

Live WTI price chart with configurable date range (1W / 1M / 3M / 1Y / All). Hover for exact values.

5.4 News Panel (■ width)

Shows the five most recent crude-market headlines with sentiment badges (Positive / Negative / Neutral) and timestamps. Click a headline to open the full article.

5.5 Alerts Panel (■ width)

Displays the most recent triggered alerts (price threshold, sentiment spike) with severity colour coding. Critical alerts appear in red, warning in amber, and informational in blue.

6. Intelligence Feed (/signals)

The Intelligence Feed aggregates all ingested news headlines and their AI-enriched metadata. It is the primary tool for analysts reviewing geopolitical and market news.

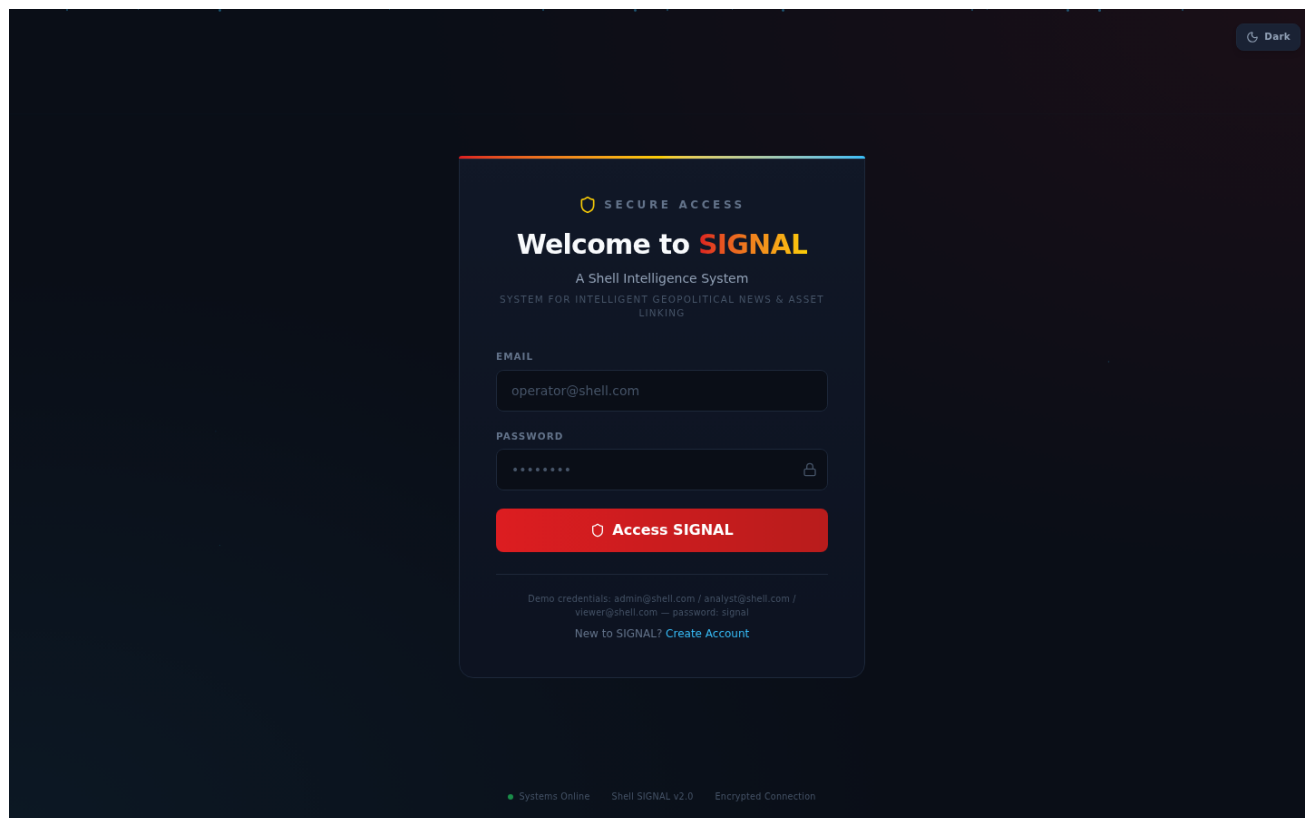


Fig 6-1 — Intelligence Feed

- **Headline table** — sortable by date, sentiment score, and source.
- **Sentiment badge** — colour-coded pill (Positive/Negative/Neutral) per article.
- **Search bar** — full-text search across all stored headlines.
- **Filter chips** — filter by sentiment or date range.
- **AI Signal indicator** — shows which articles drove the current directional prediction.

7. Commodity View (/commodity)

The Commodity View provides a detailed technical and fundamental look at crude oil prices.

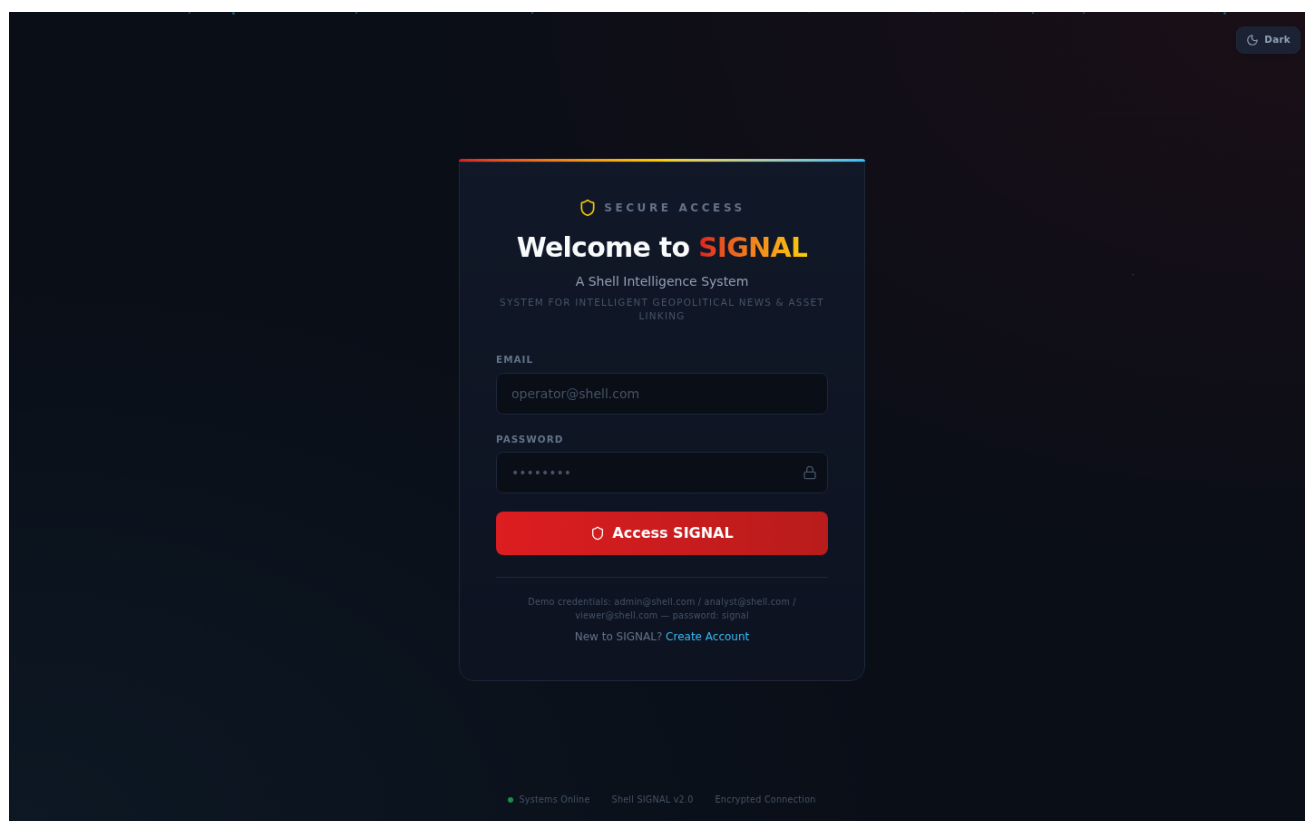


Fig 7-1 — Commodity View

- **Price chart** — full-history WTI chart with zoom/pan and date-range selector.
- **Volume bars** — trading volume overlay (where data is available).
- **Moving average overlays** — 7-day, 30-day, 90-day MAs.
- **Summary stats** — 52-week high/low, YTD performance, average daily range.

8. Analytics (/analytics)

The Analytics page surfaces sentiment and volume trends over time, helping analysts understand how the news flow is changing and what themes are driving market moves.

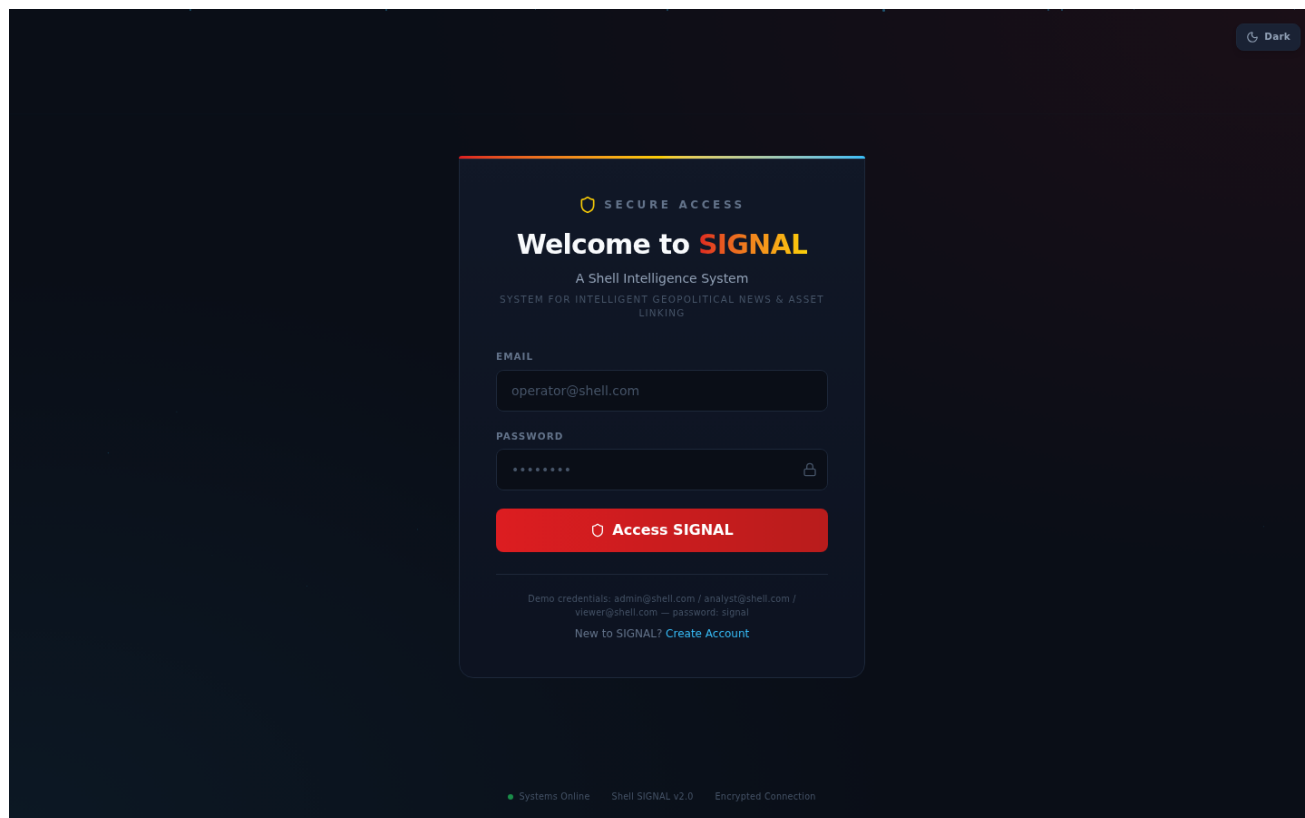


Fig 8-1 — Analytics Page

- **Sentiment trend chart** — daily average sentiment over 30 / 90 / 365 days.
- **Article volume chart** — number of relevant headlines ingested per day.
- **Signal confidence chart** — time-series of the model's directional confidence.
- **Top topics** — most frequently mentioned entities (countries, companies, events).

9. Direction Model (/model)

The Direction Model page explains and visualises the UP/DOWN classification model that powers the live signal. This page is accessible to Analyst and Executive roles.

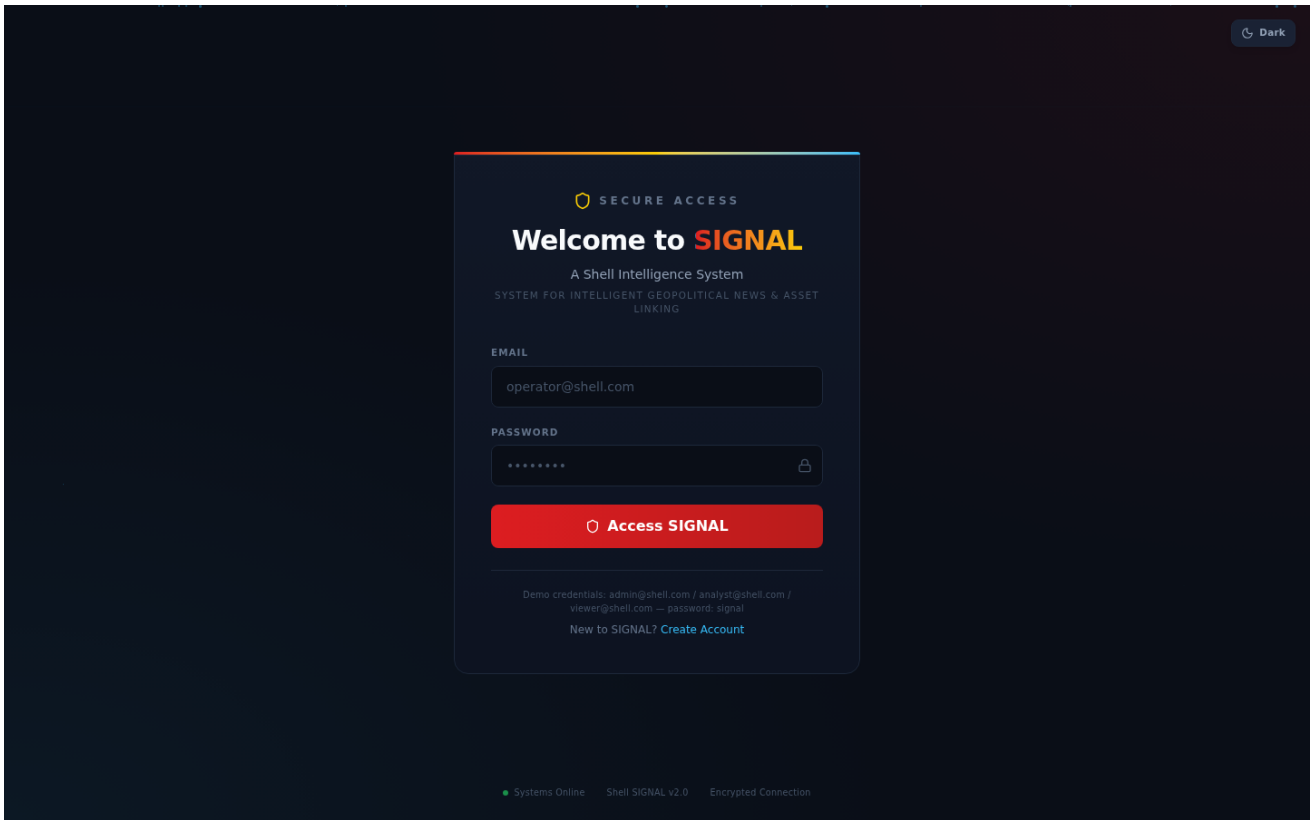


Fig 9-1 — Direction Model Page

Direction Model Metrics

Metric	Value	Interpretation
Accuracy	81.5 %	Correct UP/DOWN predictions out of all predictions
Precision	~82 %	Of all UP calls, how many were correct
Recall	~80 %	Of all actual UP days, how many were caught
F1 Score	~81 %	Harmonic mean of Precision and Recall

Prediction History Chart

A time-series chart showing daily UP/DOWN counts and average sentiment over the ingested history. Use this to spot recurring patterns (e.g., sentiment dips before DOWN days).

Feature Importance & Glossary

Bar chart of the top features driving the model's predictions. A collapsible glossary explains each feature (lag prices, rolling MAs, sentiment scores, volume metrics).

Intelligence Pipeline

A step-by-step explainer of the full data pipeline: News Ingest → NLP Enrichment → Feature Engineering → Model Inference → Signal Output.

10. Price Forecast (/forecast)

The Price Forecast page presents the regression ensemble that predicts next-day WTI closing prices. It provides everything needed to evaluate model quality and understand what is driving forecasts.

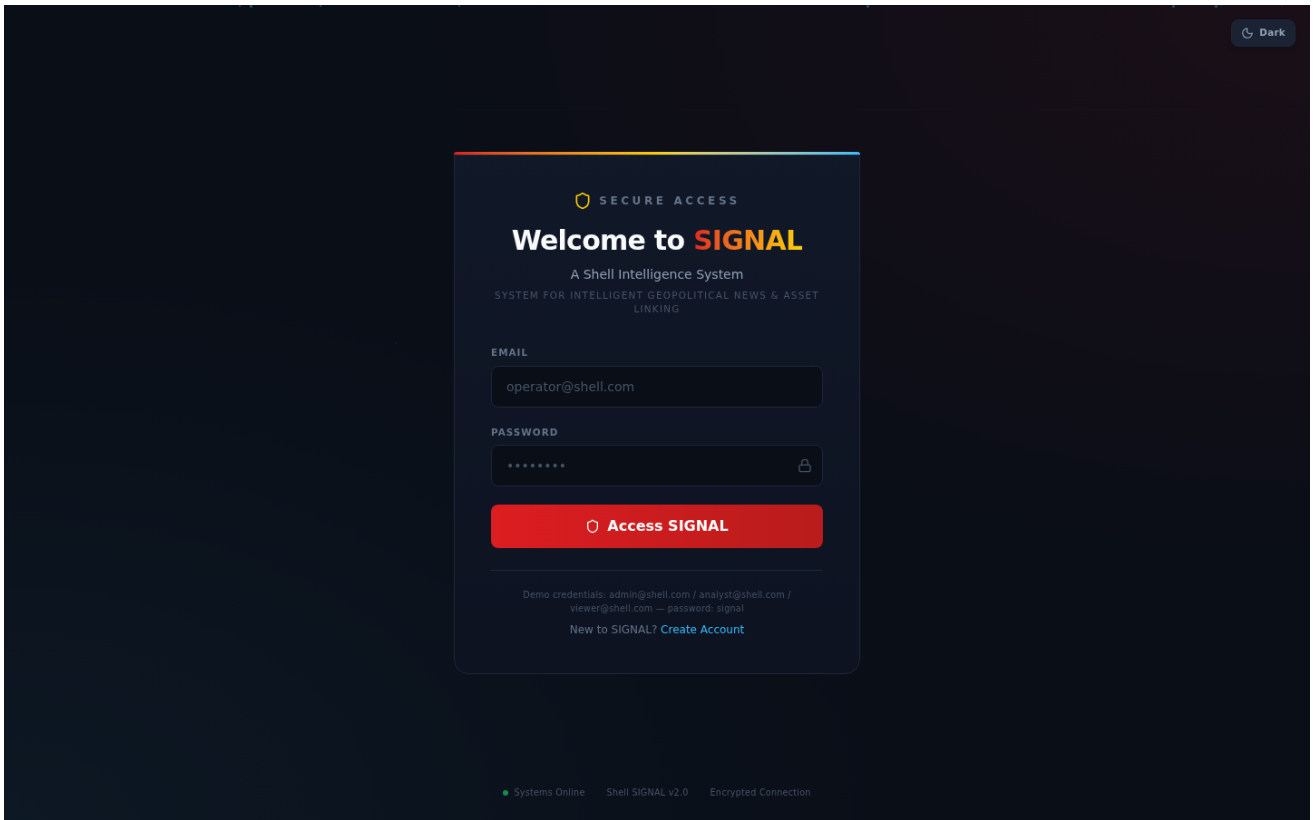


Fig 10-1 — Price Forecast Page

Performance Metrics

Metric	Value	Plain-English Meaning
RMSE	2.07 \$/bbl	Average error magnitude — model beats baseline (2.09)
MAE	~1.6 \$/bbl	Median error — typical daily miss
R ²	0.9532	95 % of price variance explained by the model
MAPE	~2.5 %	Percentage error — within 2–3 % on most days
vs Baseline	+0.7 %	Improvement over naïve last-price random-walk

Actual vs. Predicted Chart

Three-line interactive chart (Actual / Predicted / Baseline) over the holdout test window. Click legend labels to toggle individual lines. The date-range label shows the exact evaluation period. Hover for per-day values.

Error Analysis Row

Three summary cards show: predictions within $\pm 2\%$ of actual price; largest single-day miss; direction accuracy from the holdout set.

Feature Importance & Glossary

Horizontal bar chart colour-coded by feature category (lag / MA / volatility / sentiment). Collapsible glossary below provides plain-English definitions.

Model Architecture & Pipeline

3-step model architecture explainer (Base models $\rightarrow$ Meta-learner $\rightarrow$ Output) and a 4-step forecast pipeline (Data Prep $\rightarrow$ Feature Engineering $\rightarrow$ Inference $\rightarrow$ Post-process). Collapsible model comparison table lists all evaluated algorithms with their RMSE and R^2 scores.

11. Settings & Configuration (/settings)

The Settings page allows users to customise their SIGNAL experience without re-running onboarding.

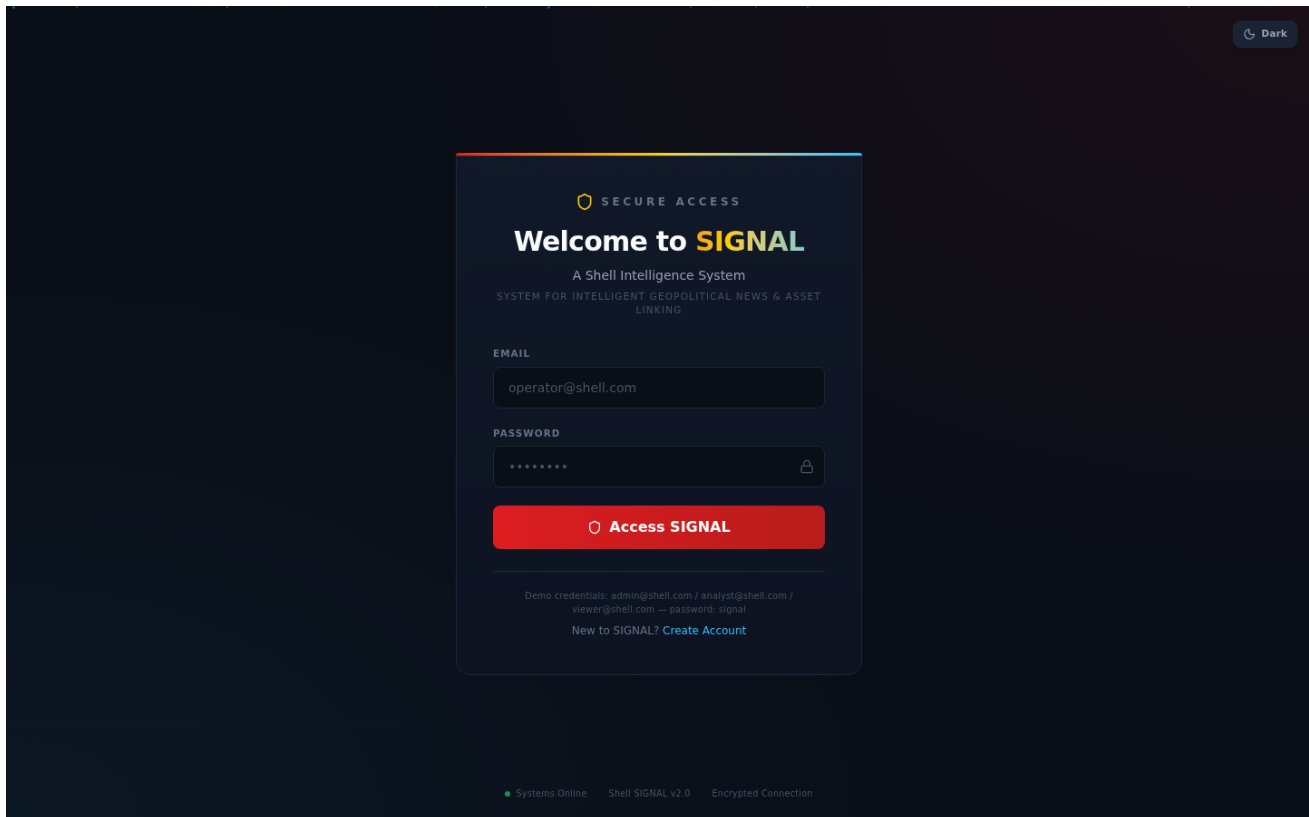


Fig 11-1 — Settings Page

11.1 Appearance

Choose between three themes using the toggle in the Settings page or the TopBar icon:

- **Dark (default)** — Dark charcoal background — optimal for extended use
- **Black** — Pure-black OLED-optimised theme
- **Light** — High-contrast light mode

11.2 Alert Configuration

Edit per-commodity price thresholds, sentiment alert levels, and notification preferences. For Executive users, also configure briefing cadence and escalation level. Changes take effect immediately without requiring a page reload.

11.3 Session

View your current session info (name, role, email) and sign out. Signing out clears all localStorage state.

12. Global Features

12.1 Command Palette (■K / Ctrl+K)

Press ■K (Mac) or **Ctrl+K** (Windows/Linux) anywhere in the dashboard to open the global Command Palette. Features:

- Navigate to any dashboard page by typing its name
- Search headlines in real time
- Run actions: Refresh Data, Logout
- Arrow-key navigation + Enter to select

12.2 Notification Center

Click the bell icon (■) in the TopBar to open the Notification Center. It shows the full alert history, severity-coded (Critical / Warning / Info). Use **Mark as Read** per alert or **Clear All** to dismiss.

12.3 Theme System

The theme toggle in the TopBar cycles through Dark → Black → Light. Your preference is persisted to localStorage key `signal_theme` and restored on every page load. You can also change themes from Settings → Appearance.

13. System Architecture

Layer	Technology	Role
Frontend	Next.js 15 (App Router) + Tailwind CSS + Recharts + Framer Motion	UI rendering, client-side auth, state management
Backend	FastAPI + SQLAlchemy + SQLite	REST API, ML inference, news query
ML Models	scikit-learn Stacking (Ridge + HistGBM + LightGBM + RidgeCV)	Price regression; RandomForest+XGBoost direction
Data	Yahoo Finance (price), NewsAPI / Reuters (headlines)	Raw data sources
Auth	localStorage JWT-like tokens (no server session)	User identity, role enforcement
Theme	CSS custom properties + ThemeProvider	Per-user appearance

Startup Commands

```
$ pip install -r requirements.txt
```

→ Install Python backend dependencies

```
$ uvicorn backend.main:app --host 0.0.0.0 --port 8000
```

→ Start FastAPI backend

```
$ cd frontend && npm install && npm run dev
```

→ Start Next.js development server (port 3000)

```
$ python train_price_model.py
```

→ Re-train the price forecast model

```
$ python train_model.py
```

→ Re-train the direction model

14. Troubleshooting

Symptom	Likely Cause	Fix
'Backend offline' banner appears	FastAPI server not running	Run: <code>uvicorn backend.main:app --port 8000</code>
Charts show no data	Backend running but DB empty	Run the data ingestion scripts: <code>python ingest_news.py</code>
Login succeeds but dashboard shows 'Unauthorized'	Role not set during registration	Re-register and select a role
Onboarding is skipped for a new account	Stale localStorage from previous session	Clear localStorage or use Incognito window
Price forecast page shows placeholder data	<code>model_report.json</code> missing or outdated	Run: <code>python train_price_model.py</code>
'TypeError: fromisoformat' in backend logs	SQLite <code>date()</code> function type mismatch	Backend patch required — see <code>backend/main.py:199-208</code>
Theme resets to dark on each page	localStorage blocked by browser policy	Allow localStorage in browser settings
Notification dropdown overlapped by chart	z-index stacking context conflict	Ensure dashboard layout main has 'relative z-0'